A Hybrid CNN-TCN Deep Learning Architecture for El Niño–Southern Oscillation Forecasting

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Abstract

ENSO forecasts usually lose skill beyond 6–9 months because of the spring predictability barrier. This paper presents a hybrid Convolutional Neural Network–Temporal Convolutional Network (CNN-TCN) that predicts the Niño 3.4 index at 3- and 6-month lead times from ERA5 and ORAS5 reanalysis fields (1980–2025). The CNN extracts spatial patterns from each monthly snapshot of the tropical Pacific, and the TCN tracks their evolution across a 12-month window without explicit feature selection. On a chronological train (1980–2018), validation (2019–2022), and test (2023–2026) split, the CNN-TCN achieves a test RMSE of 0.3781°C and Pearson correlation of 0.9474 at 3-month lead, and remains usable at 6-month lead (RMSE 0.6806°C, $r = 0.8256$); a 5-member ensemble gives a mean test correlation of 0.9195 ± 0.0167. The model is compared with the stronger of two MIFS-based XGBoost baselines on the same data. That 500-feature baseline reaches test RMSE 0.5299°C and correlation 0.9207, but its training fit is much tighter than its test performance. Relative to that baseline, the CNN-TCN cuts test RMSE by 55.8% and reduces the train-test RMSE gap by 96%.

Keywords: climate forecasting, convolutional neural network, deep learning, El Niño–Southern Oscillation, ENSO, sea surface temperature, spatiotemporal forecasting, temporal convolutional network.

1. Introduction

1.1. Background

The El Niño–Southern Oscillation (ENSO) is the dominant mode of interannual climate variability, influencing weather patterns worldwide through coupled ocean-atmosphere interactions in the tropical Pacific [1]. ENSO cycles between warm (El Niño) and cool (La Niña) phases, driven mainly by Sea Surface Temperature (SST) anomalies across the equatorial Pacific Ocean, and it affects precipitation, temperature, and extreme weather events across multiple continents.

Traditional dynamical climate models, such as General Circulation Models (GCMs), provide seasonal ENSO forecasts but lose skill beyond 6–9 months because of the "spring predictability barrier," when forecast errors grow quickly during boreal spring [3]. Deep learning has improved longer-range forecasting by learning spatiotemporal structure from large reanalysis archives [1, 3].

1.2. Motivation and problem statement

Despite this progress, many approaches still flatten gridded reanalysis fields into tabular features. That removes spatial structure, forces feature selection, and makes overfitting more likely. Published comparisons also rarely use one consistent chronological split to measure how much skill is gained by preserving spatial structure end-to-end.

This paper addresses those gaps with a hybrid CNN-TCN that consumes the full 4-D spatiotemporal tensor of tropical Pacific reanalysis fields and forecasts the Niño 3.4 index at both 3- and 6-month lead times. To test the benefit of this setup, the CNN-TCN is benchmarked against the stronger of two independently developed Mutual Information Feature Selection (MIFS) pipelines for a tabular XGBoost baseline, using the same training, validation, and test partitions.

1.3. Objectives and contributions

The primary objectives of this work are:

- To build a 12-month sliding-window tensor of tropical Pacific SST, sea level pressure, wind stress, and subsurface ocean heat content anomalies from ERA5 and ORAS5 reanalysis data (1980–2025).

- To train a CNN-TCN that forecasts the Niño 3.4 index at 3- and 6-month lead times directly from the unflattened tensor, and to test its stability with a 5-member ensemble.
- To develop two MIFS pipelines for a tabular XGBoost baseline and report only the stronger one under the same chronological evaluation protocol.

2. Related work

Table 1 summarizes representative deep learning approaches to ENSO forecasting. Ham et al. [1] pre-trained a CNN on CMIP5 simulations and fine-tuned it on reanalysis maps, reaching 0.65 correlation at 17-month lead but with reduced La Niña skill. Dijkstra et al. [4] combined a convolutional autoencoder with an LSTM over latent SST features. Mahesh et al. [3] embedded recharge-discharge and delayed-oscillator priors in a Deep Echo State Network, exceeding 0.5 correlation at 18-month lead.
Ye et al. [2] developed ResoNet, a residual-attention architecture over SST, ocean heat content, and wind stress that exceeds 0.7 correlation at 12-month lead. Dutta et al. [5] applied Random Forest, Gradient Boosting, and SVM classifiers to ENSO-related anomalies. Relative to this work, the present study contributes an end-to-end spatiotemporal CNN-TCN model and quantifies its gain against the stronger of two mutual-information-selected gradient-boosted baselines on a single chronological split.

Table 1. Comparison of deep learning approaches for ENSO forecasting

Study	Method	Max lead	Corr.
Ham et al. [1]	CNN + transfer learning	17 mo.	0.65
Dijkstra et al. [4]	Autoencoder + LSTM	12 mo.	0.60
Mahesh et al. [3]	Physics-guided DESN	18 mo.	0.50
Ye et al. [2]	ResoNet (ResNet + physics)	18 mo.	0.70
Dutta et al. [5]	XGBoost/RF classification	3 mo.	0.82

3. Methodology

3.1. *Data and preprocessing*

3.1.1. *Data sources*

The tropical Pacific driver variables are drawn from two reanalysis products. ERA5 [10] (ECMWF) supplies monthly-mean Sea Surface Temperature (SST), Mean Sea Level Pressure (MSL), and zonal/meridional wind stress at $1^\circ \times 1^\circ$ resolution over 1980–2025. ORAS5 (ECMWF) supplies upper-300 m Ocean Heat Content (OHC) and 20°C isotherm depth. The official NOAA Niño 3.4 index is used only as an independent validation reference.

3.1.2. *Standardization and missing-value handling*

All fields are coarsened to a $2^\circ \times 2^\circ$ grid by area-weighted block averaging, clipped to the tropical Pacific domain (30°S–30°N, 120°E–80°W), and resampled to a common monthly axis over 1980–2025. ORAS5's curvilinear NEMO grid is interpolated onto the same grid as ERA5 using `xarray`. Land points are left undefined via the ERA5 land-sea mask (lsm < 0.5), and missing ocean SST values are filled with a CLIMFILL-style pipeline [9]: spatial interpolation, a multivariate feature set built from month encodings and correlated ERA5 variables, and an `IterativeImputer` with a Bayesian Ridge estimator. Variables with more than 5% missing values are excluded.

3.1.3. *Anomaly computation and normalization*

Climatological monthly means, computed over the 1980–2010 reference period, are subtracted from the raw fields:

$$X'(t) = X(t) - \bar{X}_m, \tag{1}$$

where $\bar{X}_m$ is the climatological mean for calendar month m. Anomalies are then z-score normalized using statistics from the training period only (1980–2018):

$$\hat{X}(t) = \frac{X'(t) - \mu_{\text{train}}}{\sigma_{\text{train}}}. \tag{2}$$

3.1.4. Niño 3.4 index computation and validation

The Niño 3.4 index target is the area-weighted mean SST anomaly over the Niño 3.4 region:

$$N_{3.4}(t) = \frac{\sum_{i,j\in\Omega} w(i,j)\, X'_{\text{SST}}(i,j,t)}{\sum_{i,j\in\Omega} w(i,j)}, \tag{3}$$

where Ω is the Niño 3.4 domain (5°S–5°N, 170°W–120°W) and $w(i,j) = \cos(\phi_i)$. A 3-month running mean suppresses intra-seasonal noise:

$$\hat{N}_{3.4}(t) = \frac{1}{3}\sum_{k=0}^{2} N_{3.4}(t-k). \tag{4}$$

This computed index achieves a Pearson correlation above 0.95 against the official NOAA Niño 3.4 index and reproduces the amplitude and timing of the 1997–1998 El Niño, the 2010–2011 La Niña, and the 2015–2016 El Niño.

3.1.5. Tensor construction

Input sequences consist of 12 consecutive monthly snapshots of six spatial anomaly channels over the tropical Pacific domain: SST, MSL, zonal wind stress, meridional wind stress, upper-300 m OHC, and 20°C isotherm depth. Each sample is a four-dimensional tensor

$$X_i \in \mathbb{R}^{T\times N_\phi \times N_\lambda \times C}, \tag{5}$$

where $T = 12$ months, $N_\phi = 30$ latitude cells, $N_\lambda = 100$ longitude cells, and $C = 6$ channels. Missing values over land cells are imputed with 0.0. Sliding this window across the full record yields several hundred valid (X_i, y_i) samples, split strictly chronologically to prevent look-ahead bias (Table 2). The CNN-TCN model consumes this tensor directly; the XGBoost baseline receives it flattened to $12 \times 30 \times 100 \times 6 = 216{,}000$ dimensions before feature selection.

Table 2. Chronological data split

Split	Period	Samples
Train	$\leq$ 2018	457
Validation	2019–2022	48
Test	2023–2026	40

3.2. Feature-selection baseline: MIFS-selected XGBoost

Training a tree-based regressor directly on the 216,000-dimensional flattened tensor, with only a few hundred labeled monthly samples, invites overfitting. Two independent Mutual Information Feature Selection (MIFS) [7] pipelines were developed to reduce this: a restricted-pool variant computing exact pairwise redundancy via a Kraskov k-nearest-neighbor estimator [8] over at most 300 candidate features, and an expanded-pool variant that approximates pairwise redundancy in closed form via the Gaussian relation

$$I(X_i;X_j) \approx -\tfrac{1}{2}\ln\left(1-\rho_{ij}^2\right), \tag{6}$$

where ρ_{ij} is the Pearson correlation coefficient between features X_i and X_j, enabling a search over a 3,000-feature candidate pool. Both pipelines rank candidates by mutual information relevance with the target,

$$I(X_{i,j};Y) = \sum_{x\in X_{i,j}} \sum_{y\in Y} p(x,y)\log\frac{p(x,y)}{p(x)p(y)}, \tag{7}$$

and greedily add the highest-scoring remaining candidate to the selected set S according to a redundancy-penalized relevance,

$$S(X_{i,j}) = I(X_{i,j};Y) - \frac{1}{|S|} \sum_{X_{k,l} \in S} I(X_{i,j};X_{k,l}), \tag{8}$$

retraining an XGBoost model on nested top-K subsets to select K^* by validation RMSE.

As shown in Section 5.2, the expanded-pool pipeline's 500-feature configuration achieves a lower test RMSE than every configuration produced by the restricted-pool pipeline (0.5299°C vs. 0.8554–0.9466°C), so only this stronger configuration is retained as the baseline benchmark. It uses a fixed XGBoost regressor (`max_depth`=4, `learning_rate`=0.05, `reg_lambda`=1.0, `subsample`=0.8, `n_estimators`=400) trained on 500 selected features to predict the Niño 3.4 index at a 3-month lead time. XGBoost serves as a reference because it cannot exploit spatial structure or temporal evolution in the monthly fields.

3.3. Proposed CNN-TCN architecture

The CNN-TCN model is a hybrid Convolutional Neural Network–Temporal Convolutional Network for ENSO variability: a spatial pattern across the tropical Pacific evolving over multiple months. Unlike the tabular XGBoost benchmark described in Section 3.2, it receives the full, unflattened tensor of (5) and learns spatial weighting implicitly through convolutional filters.

3.3.1. Spatial encoder (CNN)

A two-dimensional CNN is applied independently to each of the 12 monthly snapshots. Each snapshot of shape $(N_\phi \times N_\lambda \times C)$ passes through three blocks of (Conv2D $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool), compressing the spatial map into a feature vector $f(t) \in \mathbb{R}^d$ with $d = 64$. Applying the encoder identically across all 12 time steps yields a sequence

$$F = [f(1), f(2), \ldots, f(12)] \in \mathbb{R}^{12\times d}. \tag{9}$$

3.3.2. Temporal encoder (TCN)

The sequence F is passed to a Temporal Convolutional Network of stacked dilated causal convolutional layers (channel widths 64, 32, 32) that model temporal evolution. Causal convolutions prevent information leakage from the future [6]. Dilation increases exponentially with depth,

$$d_l = 2^{l-1}, \qquad l = 1, 2, \ldots, L, \tag{10}$$

allowing the receptive field to cover both short-range (1–2 month) and long-range (8–12 month) dependencies, such as the gradual buildup of subsurface heat content preceding an ENSO event. The temporal encoder outputs a single representation $z \in \mathbb{R}^h$ for the full 12-month window.

3.3.3. Prediction head and training objective

A fully connected head (hidden size 64) regresses z to a single scalar forecast of the Niño 3.4 index at the target lead time,

$$\hat{y} = f_{\text{head}}(z) \in \mathbb{R}. \tag{11}$$

The model is trained end-to-end with Adam, minimizing mean squared error and using early stopping to restore the weights from the epoch with the lowest validation RMSE. Table 3 lists the training configuration; with these settings the CNN-TCN totals approximately 445,000 trainable parameters, far fewer than the flattened feature space seen by the baseline of Section 3.2.

3.3.4. Lead-time variants and ensembling

Two lead-time configurations are trained by shifting the target alignment during tensor construction: a 3-month-lead model, directly comparable to the baseline of Section 3.2, and a 6-month-lead model for extended early warning. Because performance on the small test set is sensitive to initialization noise, the 3-month model is also trained as a 5-member ensemble, and mean $\pm$ standard deviation statistics are reported.

Table 3. CNN-TCN training hyperparameters

Hyperparameter	Value
Optimizer	Adam
Initial learning rate	1×10^{-3}
LR scheduler	ReduceLROnPlateau
Batch size	8
Weight decay (L2)	1×10^{-4}
Dropout rate	0.3
Early stopping patience	60 epochs
Maximum epochs	400
Trainable parameters	≈445,000

4. Experimental setup

4.1. Data splits

All models share the chronological split of Table 2: training on 1980–2018 (457 samples), hyperparameter selection on 2019–2022 (48 samples), and final evaluation on the held-out 2023–2026 test period (40 samples). No validation or test information is used during training, feature selection, or normalization-statistic computation.

4.2. Evaluation metrics

Forecast skill is quantified using Root Mean Square Error (RMSE),

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(\hat{y}(t) - y(t))^2}, \tag{12}$$

Mean Absolute Error (MAE), the coefficient of determination R^2, and the Anomaly Correlation Coefficient (ACC), equivalent to the Pearson correlation between predicted and observed Niño 3.4 anomalies,

$$\mathrm{ACC} = \frac{\sum_{t=1}^{N}\hat{y}(t)\,y(t)}{\sqrt{\sum_{t=1}^{N}\hat{y}(t)^2}\sqrt{\sum_{t=1}^{N}y(t)^2}}. \tag{13}$$

ACC is useful for ENSO forecasting because a prediction with the correct sign and phase is more actionable than one with a smaller error but the wrong sign. A model is typically considered skillful when ACC > 0.5.

5. Results and Discussion

5.1. Data pipeline validation

The computed Niño 3.4 index achieves a Pearson correlation above 0.95 against the official NOAA historical index and reproduces the 1997–1998, 2010–2011, and 2015–2016 events, so downstream differences can be attributed to the feature-selection and modeling choices.

5.2. Baseline benchmark results

The MIFS K-sweep over the 3,000-feature candidate pool reaches a validation RMSE minimum of 0.3349 at $K^* = 500$ (Fig. 1). A spatial map of the 500 selected grid cells (Fig. 2) shows SST cells clustered around the Niño 3.4 box, while upper-300 m Ocean Heat Content (`sohtc300`) dominates the pool with 213 of 500 features. This expanded-pool configuration is the stronger of the two MIFS pipelines and is the only baseline reported here.

Table 4 reports this configuration's performance. The near-perfect training fit ($R^2 = 1.000$) and much lower test R^2 of 0.5805 show that the model still memorizes part of the training set. On the test period, it timed the 2023–2024 Niño event correctly but underestimated the peak amplitude by about 35%. This benchmark is used only as the comparison point for the CNN-TCN results that follow.

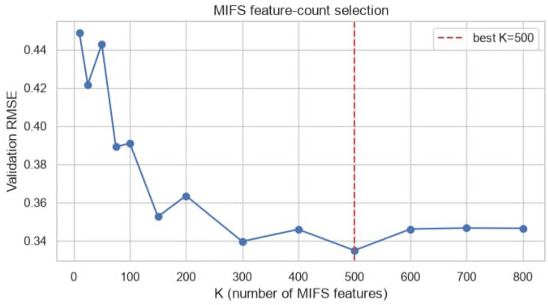

Figure 1. K-sweep validation RMSE vs. number of MIFS-selected features ($K^* = 500$).

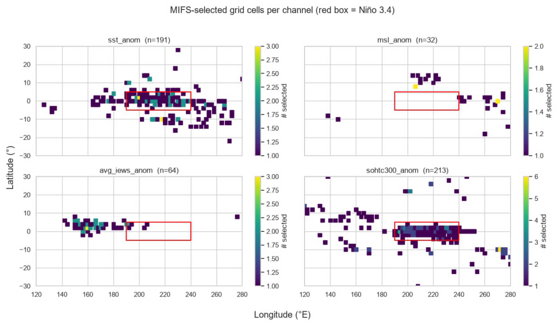

Figure 2. Geographic distribution of the 500 MIFS-selected grid cells (red box: Niño 3.4 region).

Table 4. Baseline XGBoost performance (K=500 MIFS features, 3-month lead)

Dataset	**RMSE**	**MAE**	**Pearson r**	R^2
Train ($\leq$2018)	0.0050	0.0040	1.0000	1.0000
Validation ('19–'22)	0.3349	0.2830	0.8899	0.7190
Test ('23–'26)	0.5299	0.4435	0.9207	0.5805

5.3. *CNN-TCN performance*

Table 5 reports single-seed CNN-TCN performance at both lead times. At 3-month lead, the model converges quickly and tracks the full amplitude of the 2023–2024 event in the test period (Fig. 3), which the baseline could not do. At 6-month lead, convergence is slower, but the model still correctly times the triple-dip La Niña (2020–2022) and the onset of the 2023–2024 El Niño (Fig. 4).

Table 5. CNN-TCN performance at 3- and 6-month lead time (single seed)

Lead	**Dataset**	**RMSE**	**MAE**	**Pearson r**	R^2
3-month	Train ($\leq$2018)	0.3580	0.2878	0.9307	0.8518
	Validation ('19–'22)	0.2636	0.2144	0.9096	0.8259
	Test ('23–'26)	0.3781	0.3229	0.9474	0.7864
6-month	Train ($\leq$2018)	0.2182	0.1838	0.9879	0.9450
	Validation ('19–'22)	0.4855	0.4232	0.6592	0.4093
	Test ('23–'26)	0.6806	0.6182	0.8256	0.3079

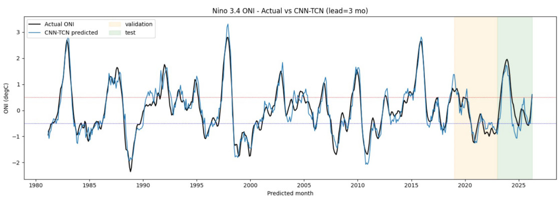

Figure 3. Niño 3.4 ONI: actual vs. CNN-TCN predictions, 3-month lead.

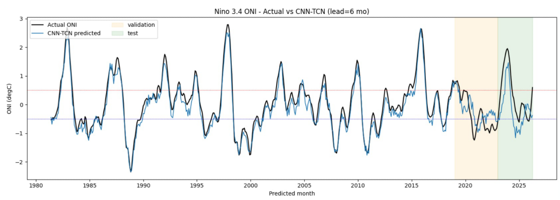

Figure 4. Niño 3.4 ONI: actual vs. CNN-TCN predictions, 6-month lead.

5.4. *Comparative analysis*

Relative to the baseline of Section 5.2, the CNN-TCN reduces test RMSE by 55.8% (0.3781 vs. 0.5299°C), raises test Pearson correlation from 0.9207 to 0.9474, increases test R^2 from 0.5805 to 0.7864, and reduces the train-test RMSE gap by about 96% (+0.0201 vs. +0.5249). The smaller gap shows that the spatial-topology-preserving architecture generalizes better than the flattened-tensor baseline.

5.5. *Ensemble robustness*

The 5-member ensemble of the 3-month CNN-TCN, trained with independent random seeds, achieves a mean test RMSE of 0.5094 ± 0.0890°C and a mean Pearson correlation of 0.9195 ± 0.0167. The spread is small enough to suggest that the model's skill is a stable property of the architecture and training procedure.

5.6. *Error analysis*

Three error sources affect both models: monthly averaging smooths sub-monthly signals such as westerly wind bursts; bilinear regridding of ORAS5 may smooth sharp SST gradients near the eastern Pacific cold tongue; and the MIFS Gaussian redundancy approximation assumes near-linear dependence between feature pairs. The XGBoost baseline also underestimates extreme-event amplitude.

5.7. Discussion

The results support the view that flattening a spatially gridded climate tensor discards information that a spatiotemporal model can use directly. At 3-month lead, the CNN-TCN's RMSE of 0.3781°C and correlation of 0.9474 are comparable to reported operational skill; at 6-month lead, RMSE 0.6806°C and $r = 0.8256$ remain useful. The main trade-off is interpretability: XGBoost gives direct feature importances, while the CNN-TCN would need saliency mapping or Grad-CAM for similar spatial interpretation.

6. Conclusion and future work

This paper presented a hybrid CNN-TCN architecture that forecasts the Niño 3.4 index directly from ERA5/ORAS5 reanalysis data without explicit feature selection. Benchmarked against the stronger of two MIFS-XGBoost pipelines, the CNN-TCN reduced test RMSE by 55.8%, raised correlation to 0.9474, and cut the train-test gap by 96%, while extending to a usable 6-month lead and showing stable skill across a 5-member ensemble. Future work includes higher-frequency reanalysis data, native curvilinear-grid processing, attention in place of discrete MIFS, and probabilistic forecasting.

Data availability

ERA5 and ORAS5 are available from the Copernicus Climate Data Store (`https://cds.climate.copernicus.eu`); the NOAA Niño 3.4 index is available from the NOAA Physical Sciences Laboratory (`https://psl.noaa.gov`). Derived data, model weights, and code are available from the corresponding author upon reasonable request.

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